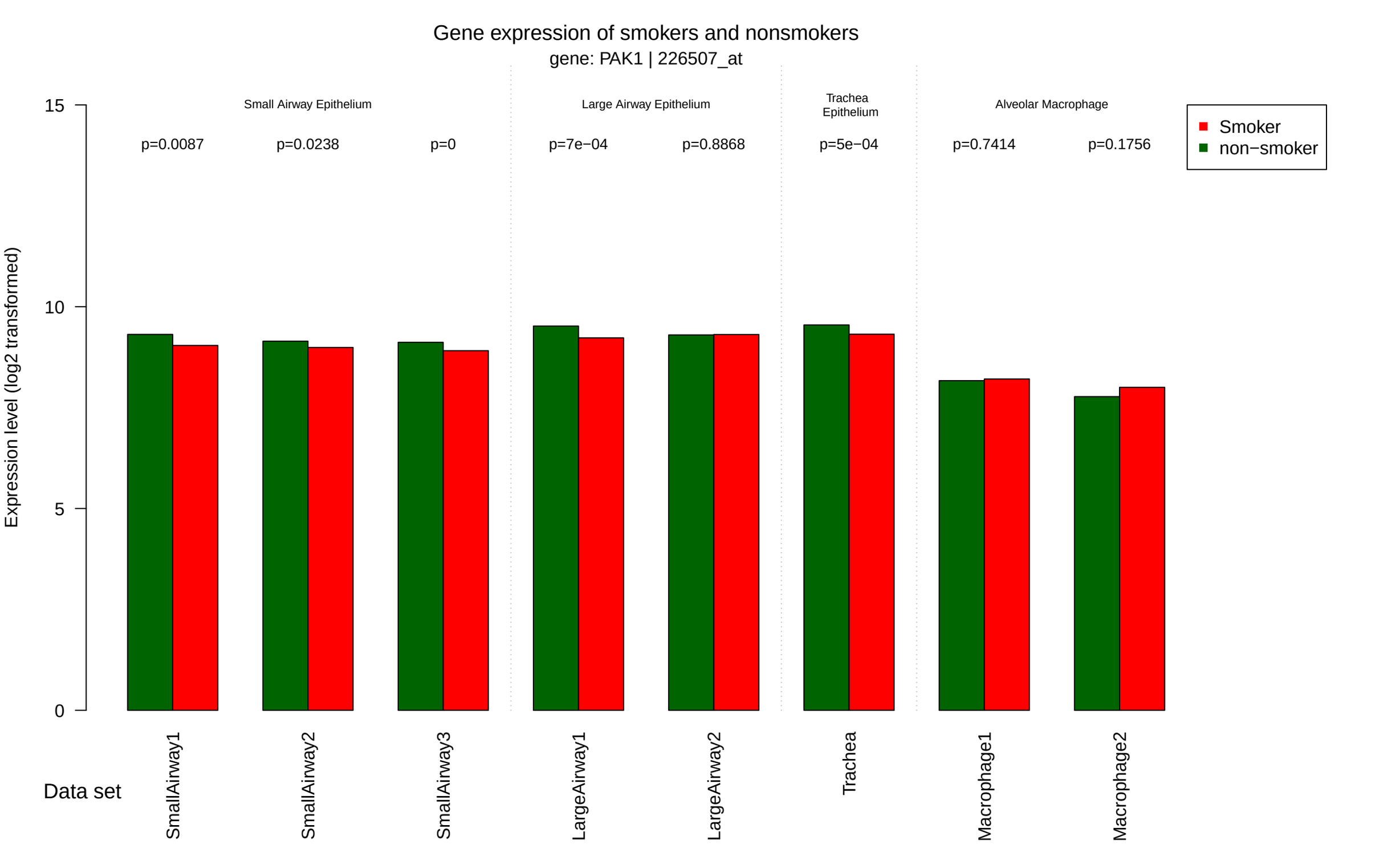
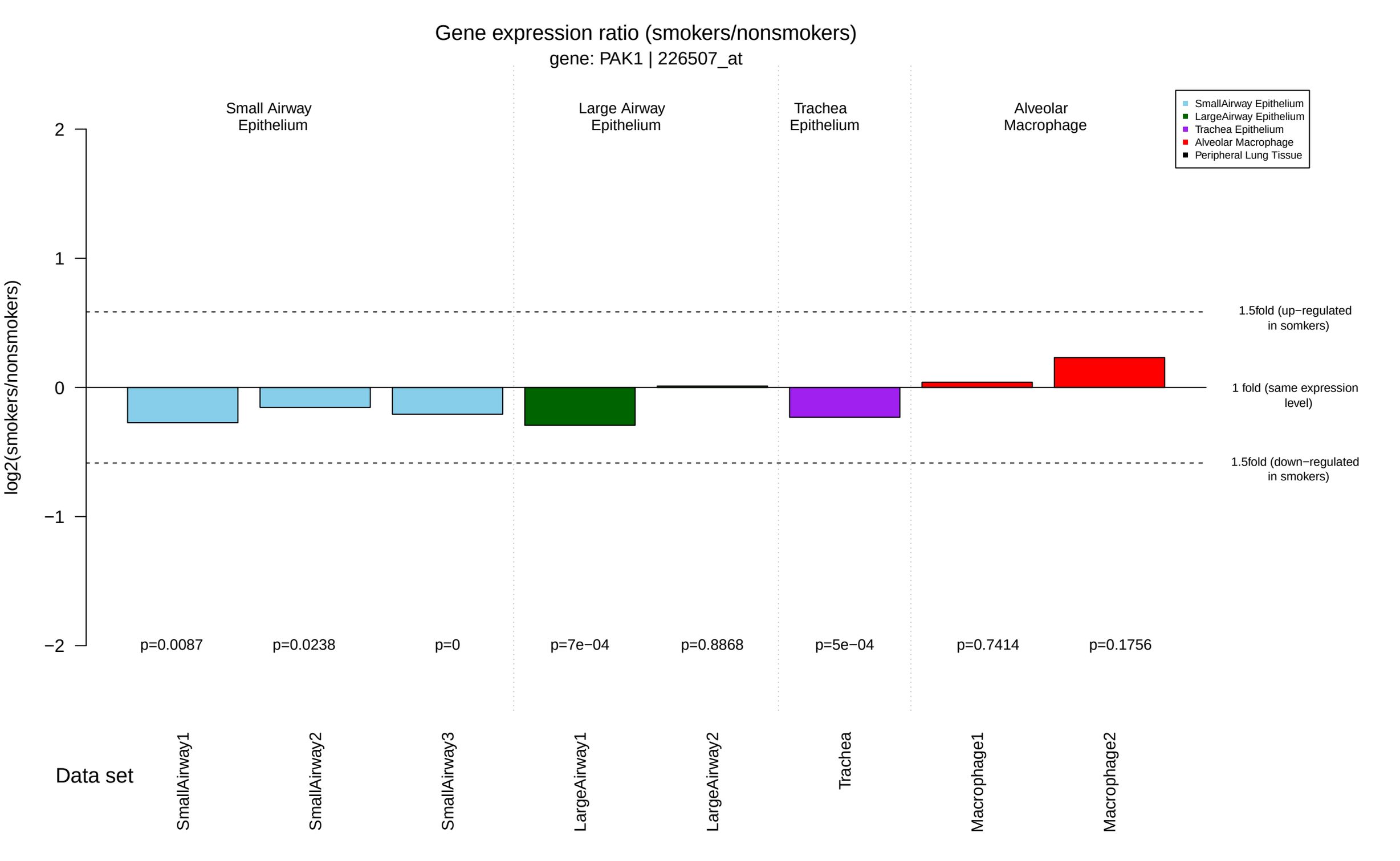
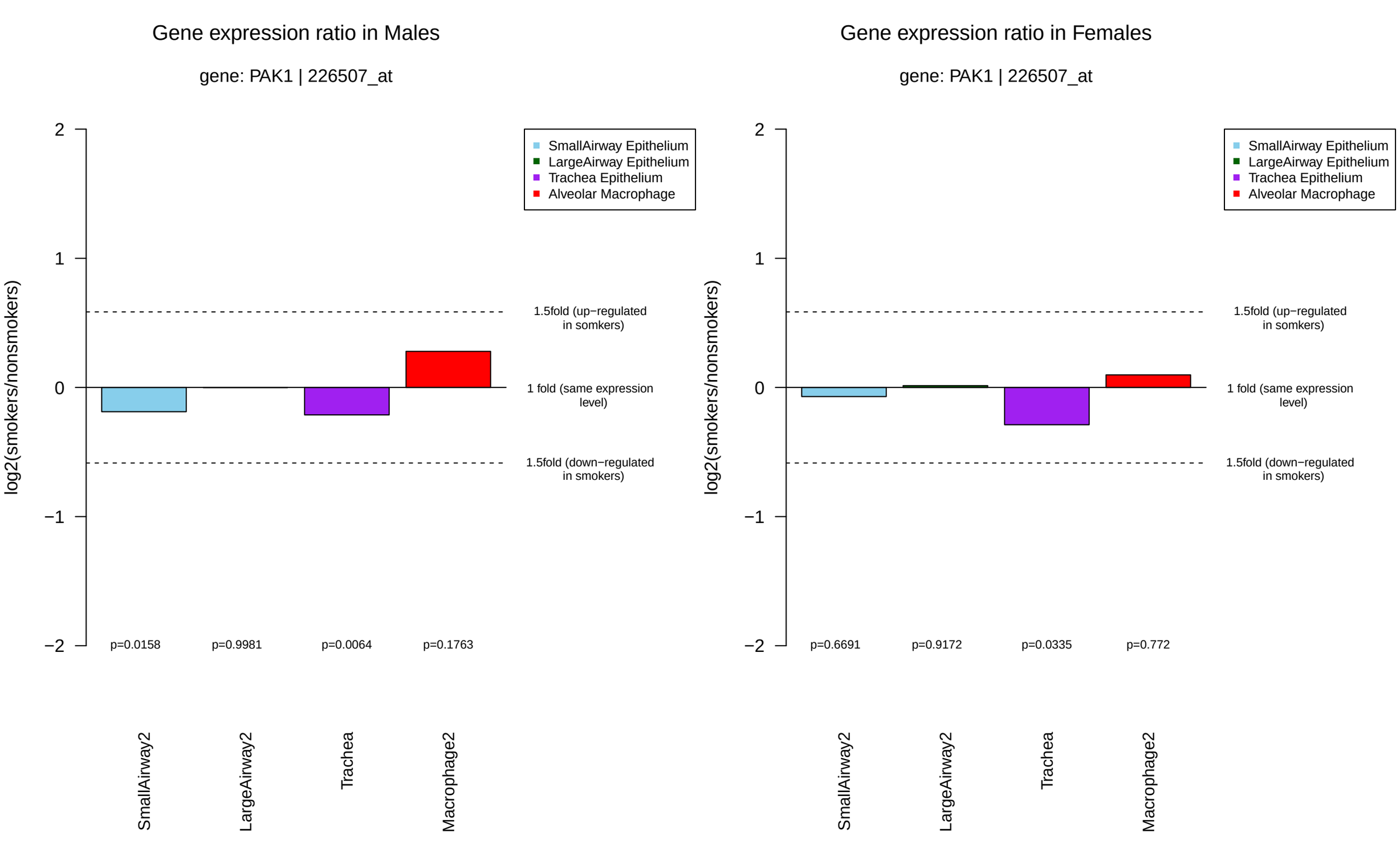
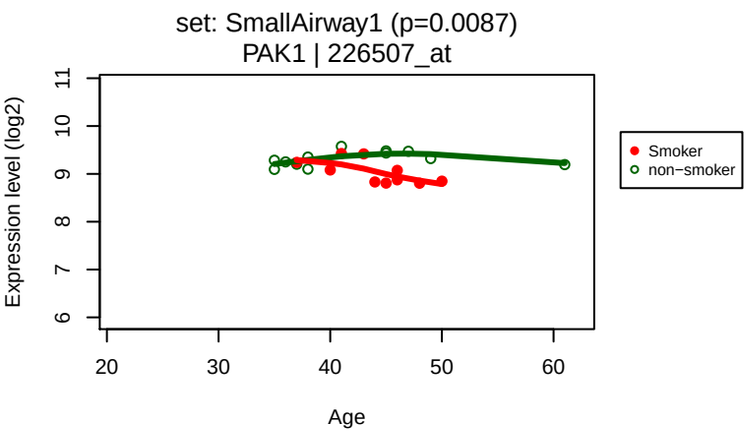
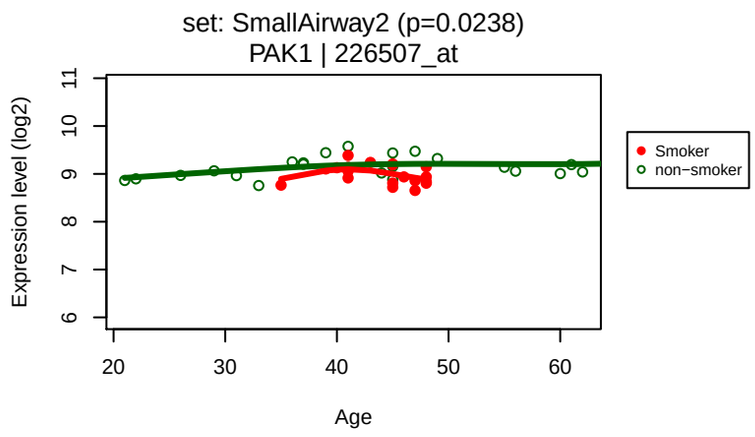
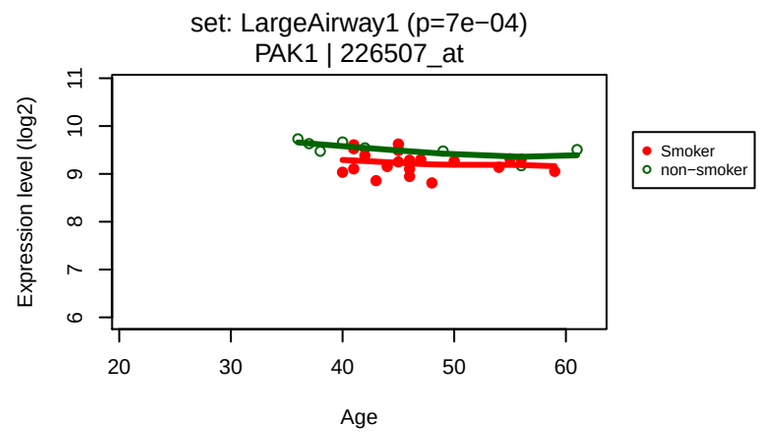

Gene expression of smokers and nonsmokers
gene: PAK1 | 226507_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0087
p=0.0238
p=0
p=7e−04
p=0.8868
p=5e−04
p=0.7414
p=0.1756
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

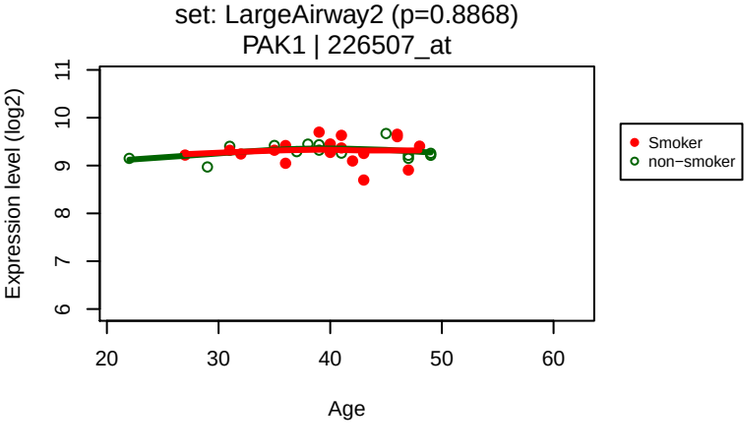

Gene expression ratio (smokers/nonsmokers)
gene: PAK1 | 226507_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0087
p=0.0238
p=0
p=7e−04
p=0.8868
p=5e−04
p=0.7414
p=0.1756
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

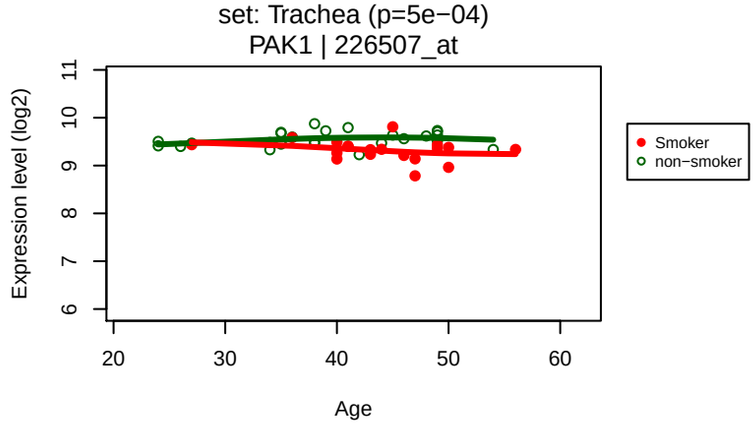

Gene expression ratio in Males
gene: PAK1 | 226507_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0158
p=0.9981
p=0.0064
p=0.1763
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAK1 | 226507_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6691
p=0.9172
p=0.0335
p=0.772
SmallAirway2
LargeAirway2
Trachea
Macrophage2

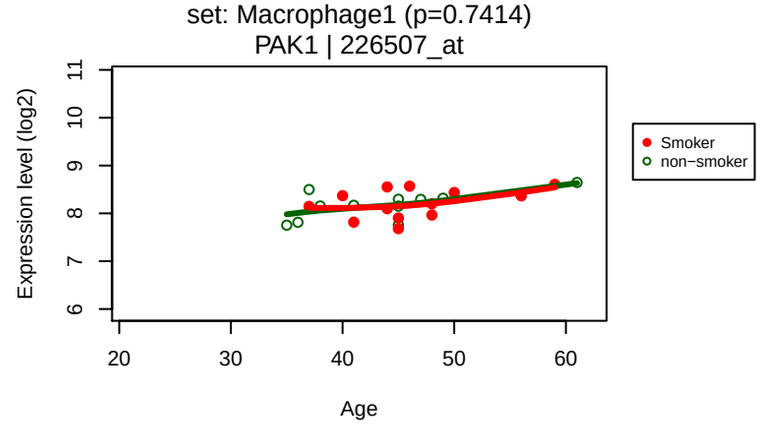

set: SmallAirway1 (p=0.0087)
PAK1 | 226507_at
Expression level (log2)
Age
Smoker
non−smoker

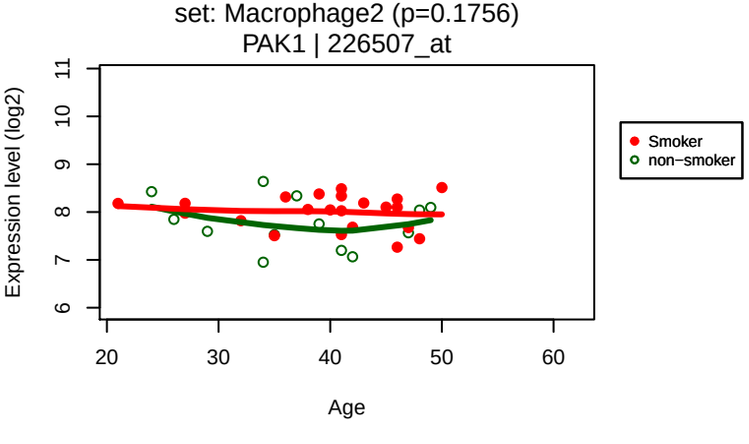

set: SmallAirway2 (p=0.0238)
PAK1 | 226507_at
Expression level (log2)
Age
Smoker
non−smoker

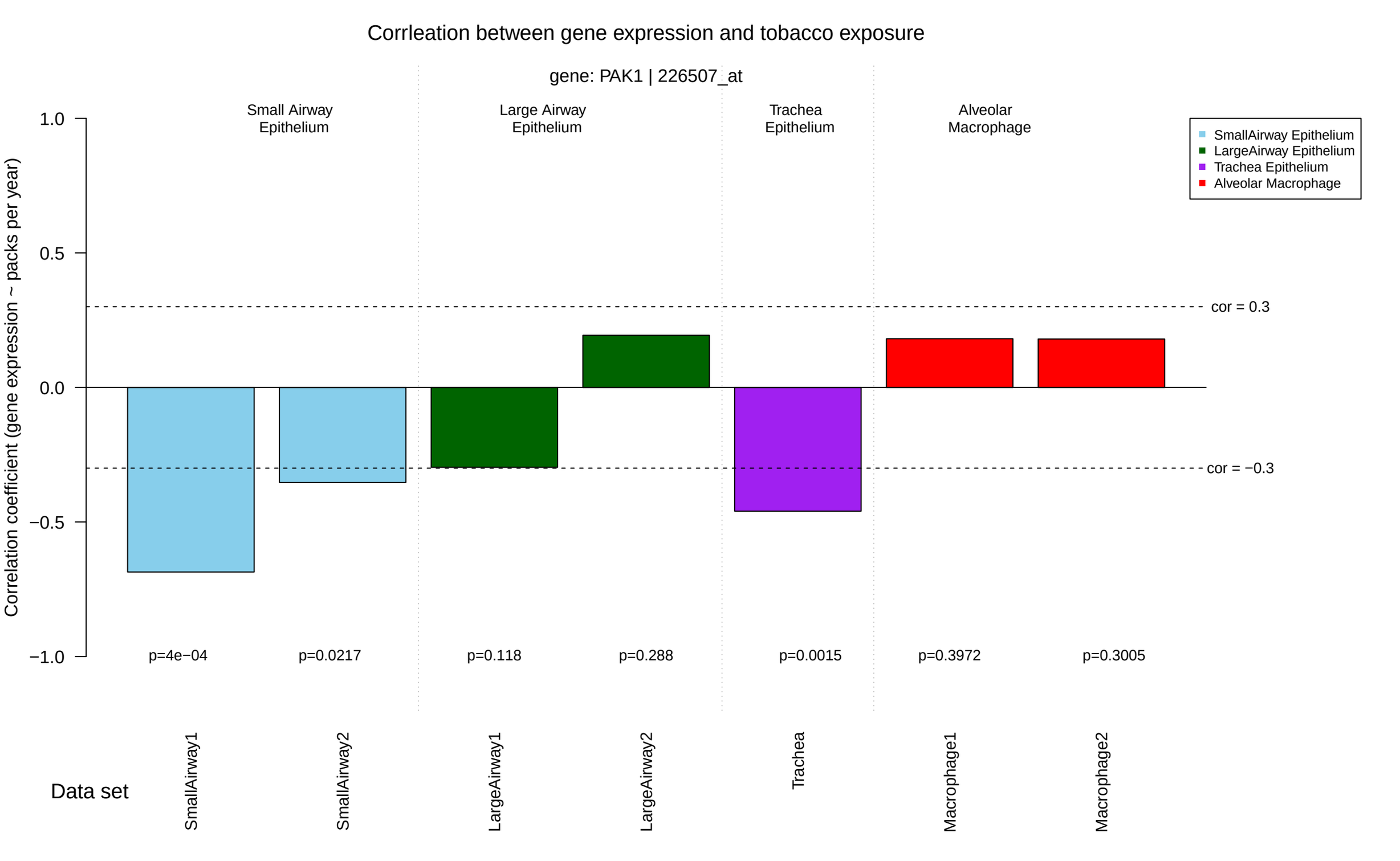

set: LargeAirway1 (p=7e−04)
PAK1 | 226507_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8868)
PAK1 | 226507_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=5e−04)
PAK1 | 226507_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7414)
PAK1 | 226507_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1756)
PAK1 | 226507_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAK1 | 226507_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=4e−04
p=0.0217
p=0.118
p=0.288
p=0.0015
p=0.3972
p=0.3005
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2